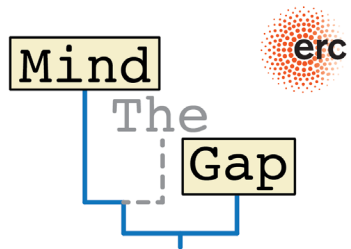


Age-depth modeling beyond absolute ages

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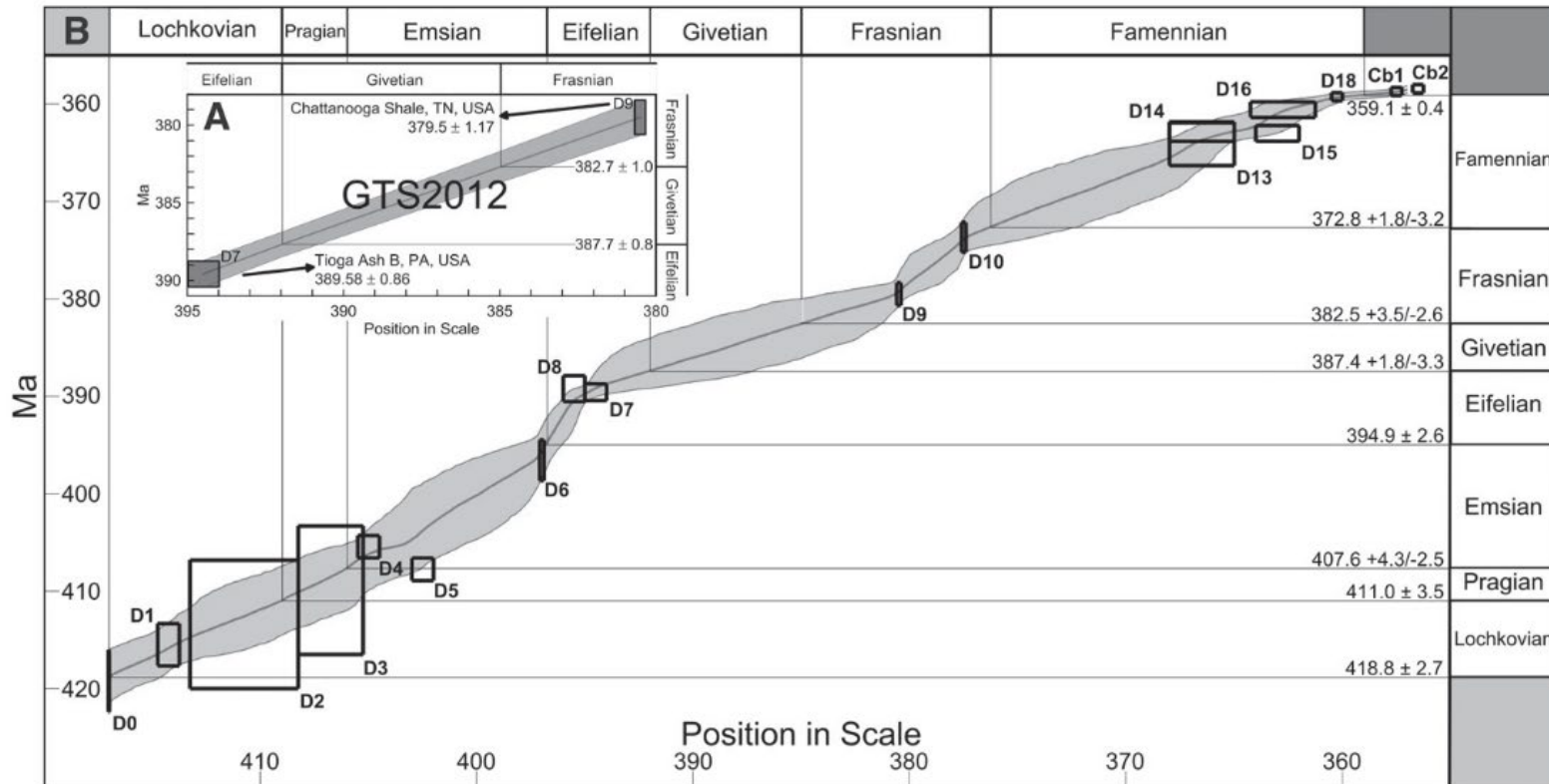
Scales of measurement

Name
Absolute scale
Interval/difference scale
Ratio scale
Ordinal scale

Scales of measurement

Name	Question	Dimension/Unit
Absolute scale	How tall are children?	L/m
Interval/difference scale	How much taller are children?	L/m
Ratio scale	Children are x times taller	None
Ordinal scale	Which children are taller?	None

Absolute scale: Absolute age-depth models

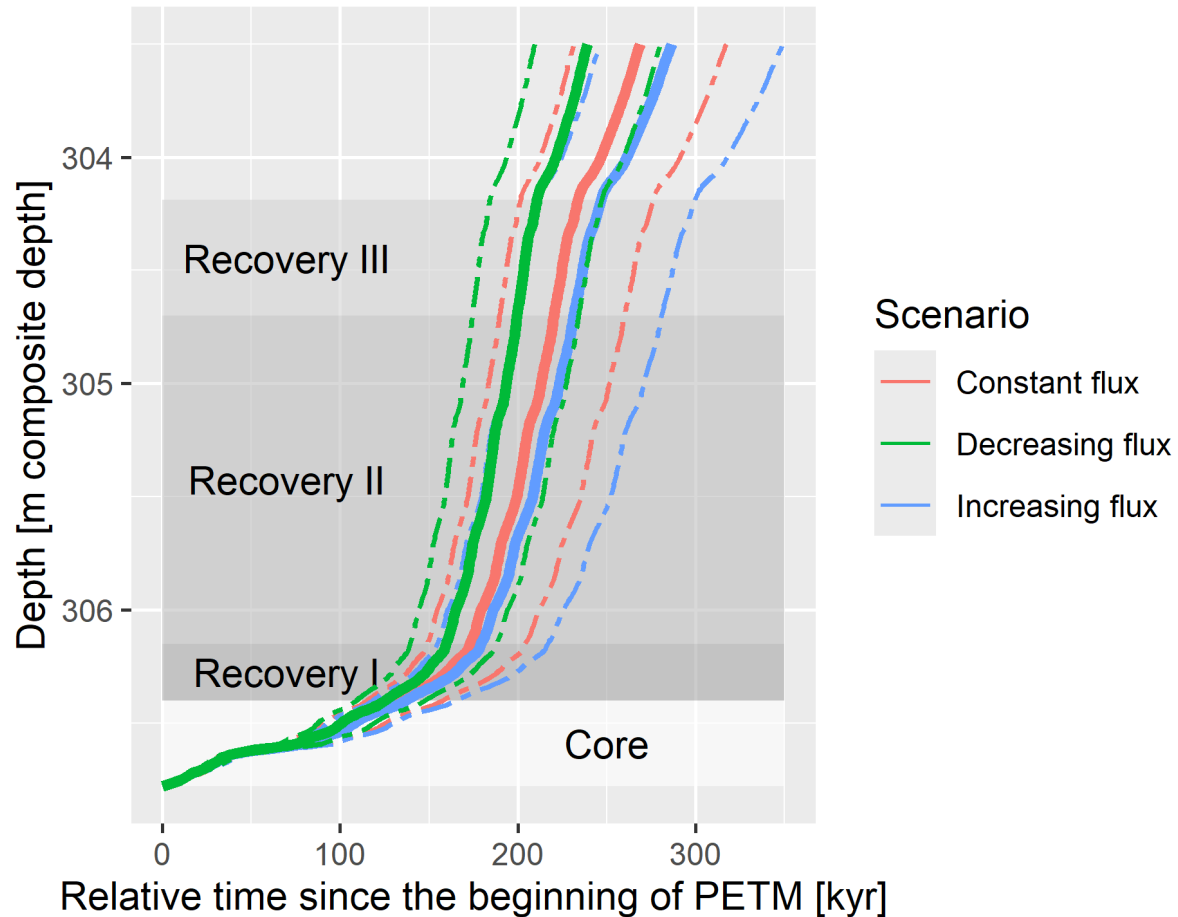


Units: **Ma**

Dominant discipline:
Geochronology (e.g.,
radioisotope systems)

Interval scale: Anchored age-depth models

Age-depth models for PETM at ODP Site 1266

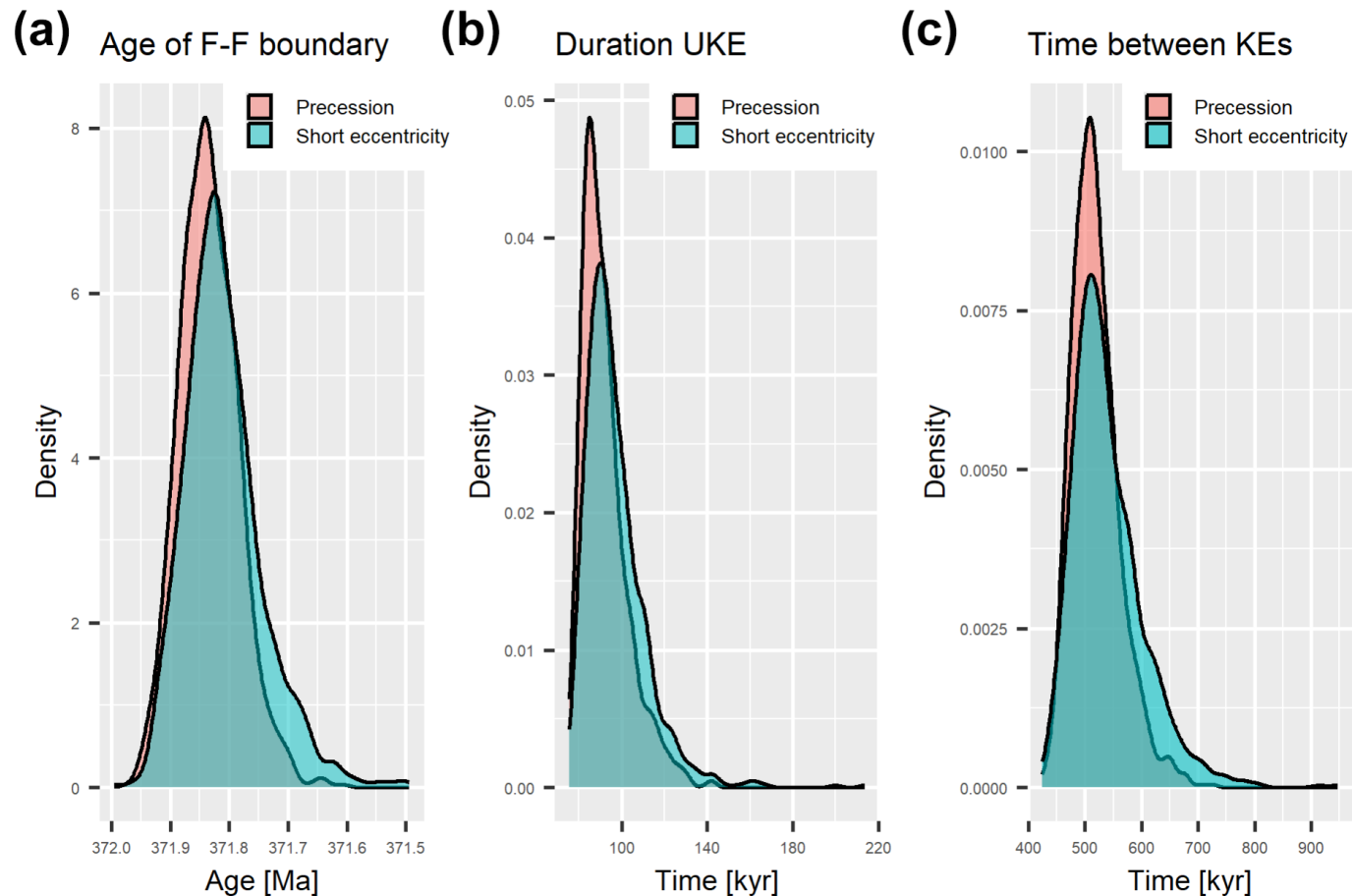


Unit: **Myr**

Measurement of time
passed relative to a
predefined horizon

Dominant source of
information: Sediment
accumulation

Different sources of uncertainty

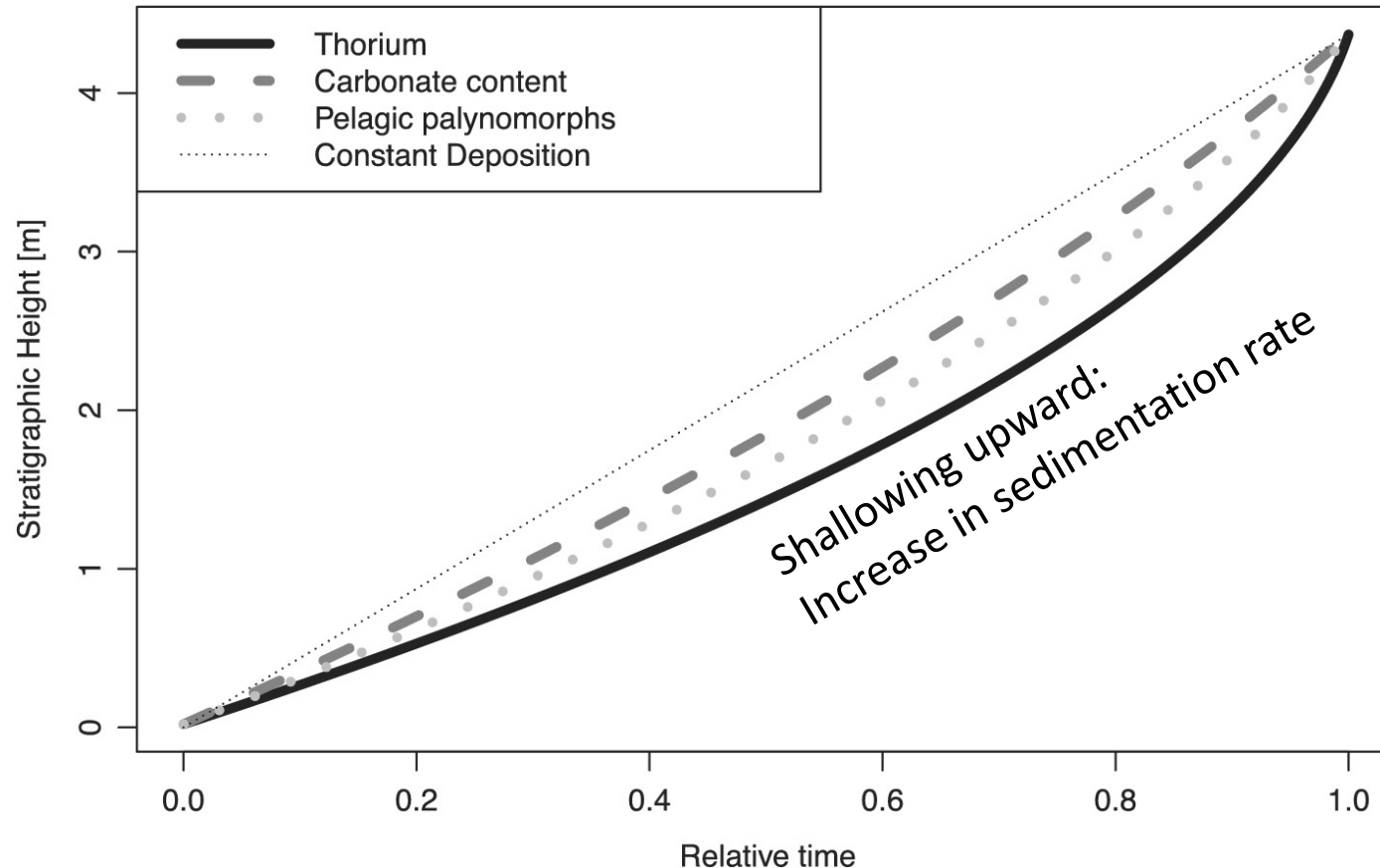


(a) Dominated by uncertainty of absolute ages (2 sigma: 50 kyr)

(b) & (c) Uncertainty from sedimentation rates/sedimentary systems dynamics

-> Can estimate durations with precisions below absolute age uncertainties with a good understanding of sedimentary systems

Ratio scale: Floating (relative) age-depth models

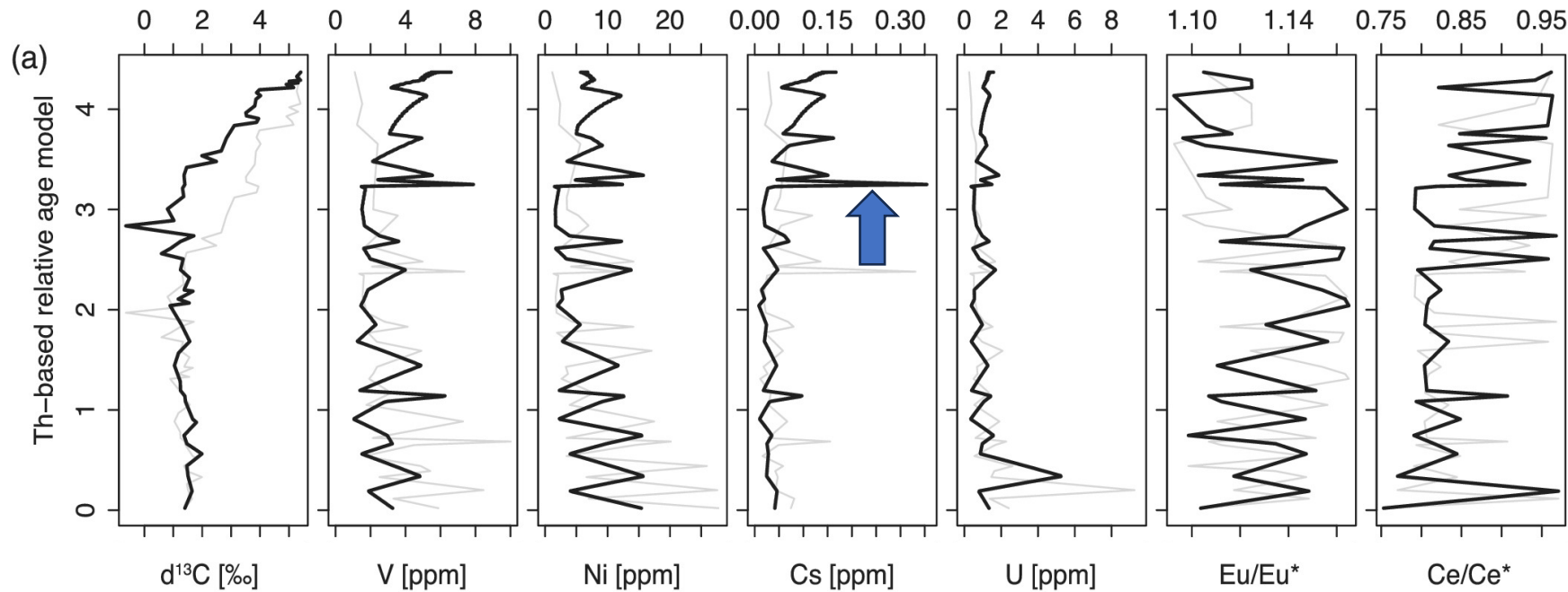


Units: **None**

Measures relative
distribution of time in section

Dominant source of
information: Relative changes
in sedimentation rates

Floating (relative) age-depth models



Correct rates
and fluxes
within section!

Black: Corrected for changing sedimentation rates

Ordinal scale: ???

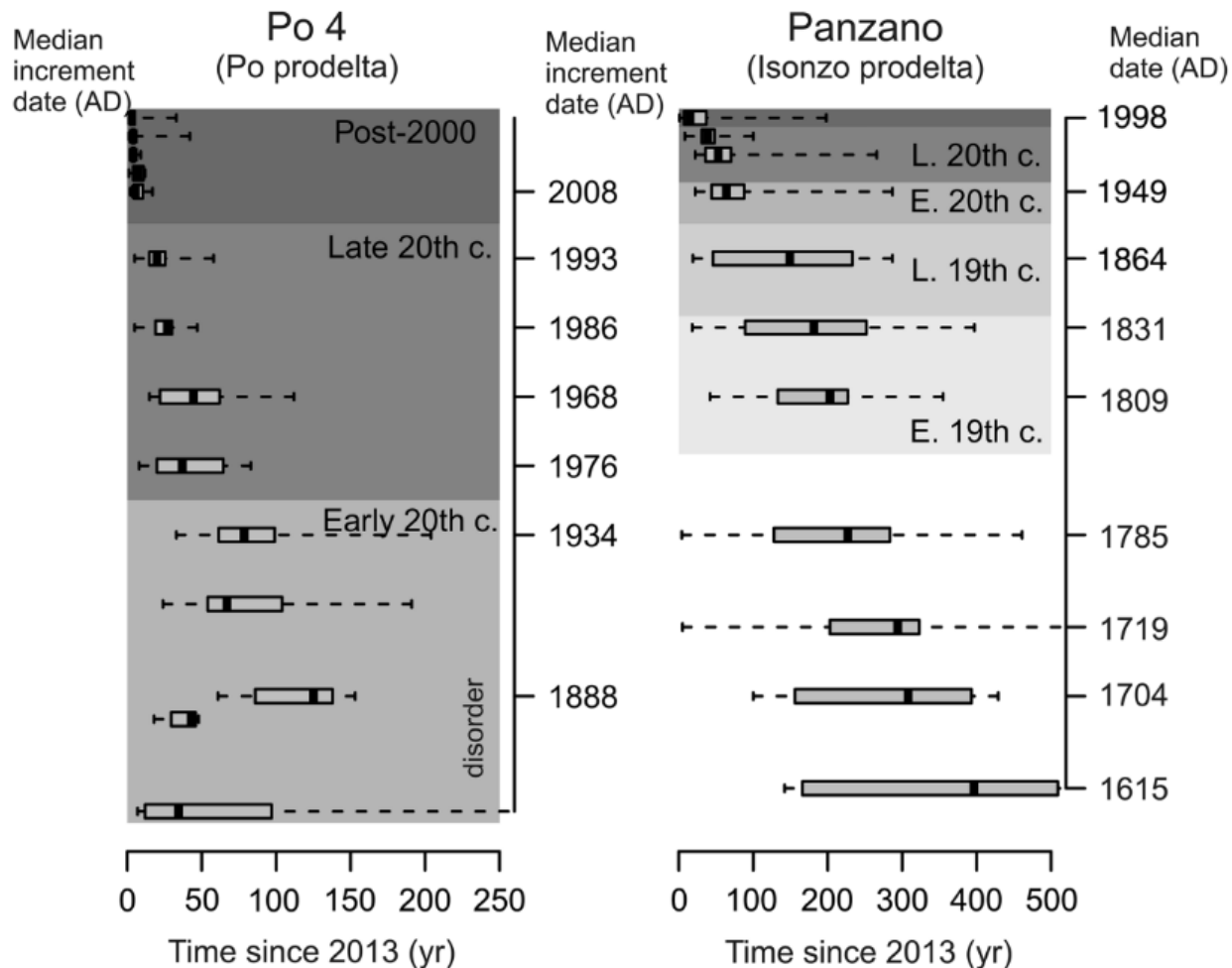
Law of superposition/Steno's law

Model assumption in all age-depth models
-> monotonicity



Nicolaus Steno

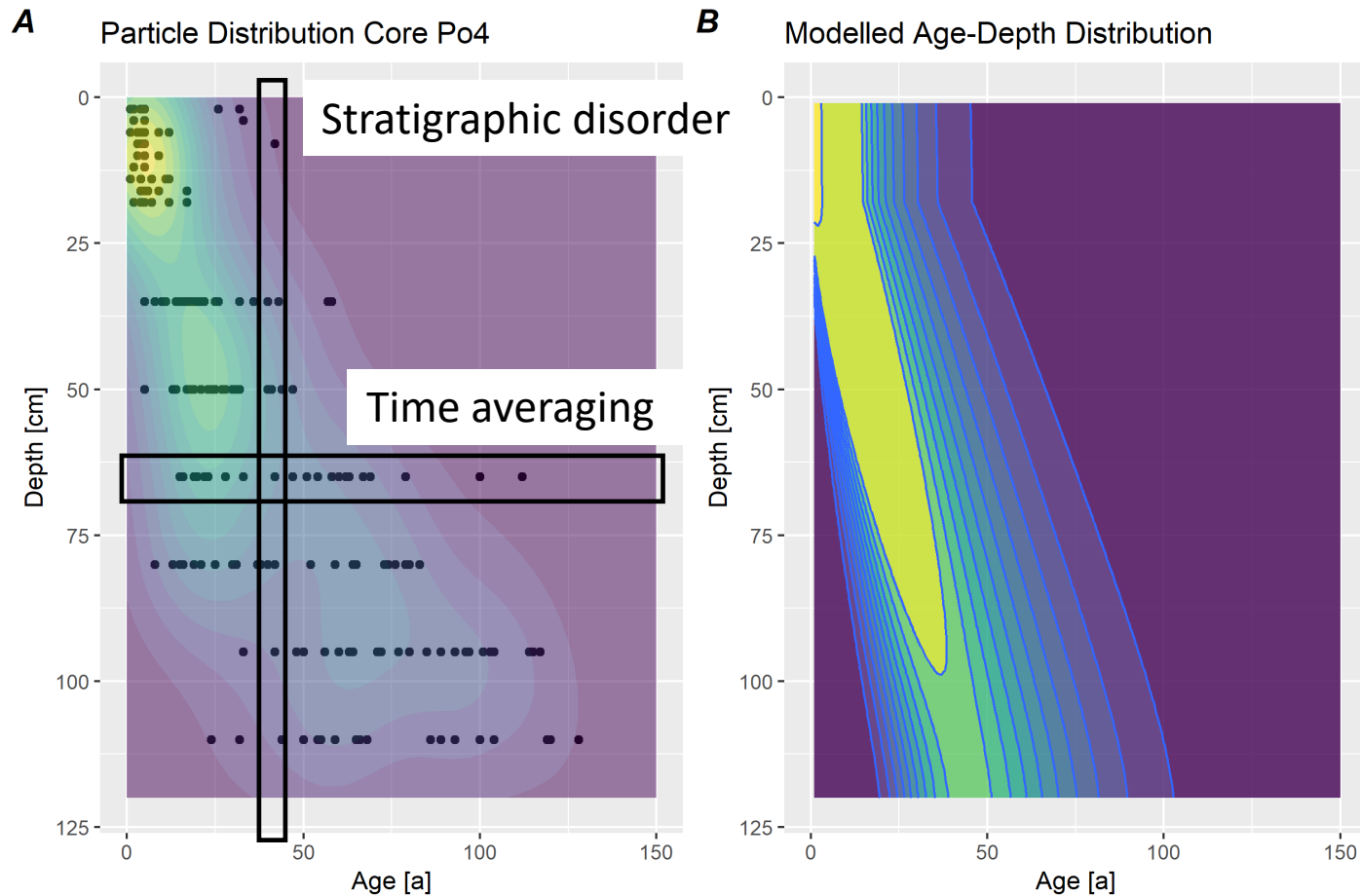
Limits of superposition



Time-averaging and stratigraphic disorder due to sediment mixing

Dominant driver:
Physics/biology of particle deposition

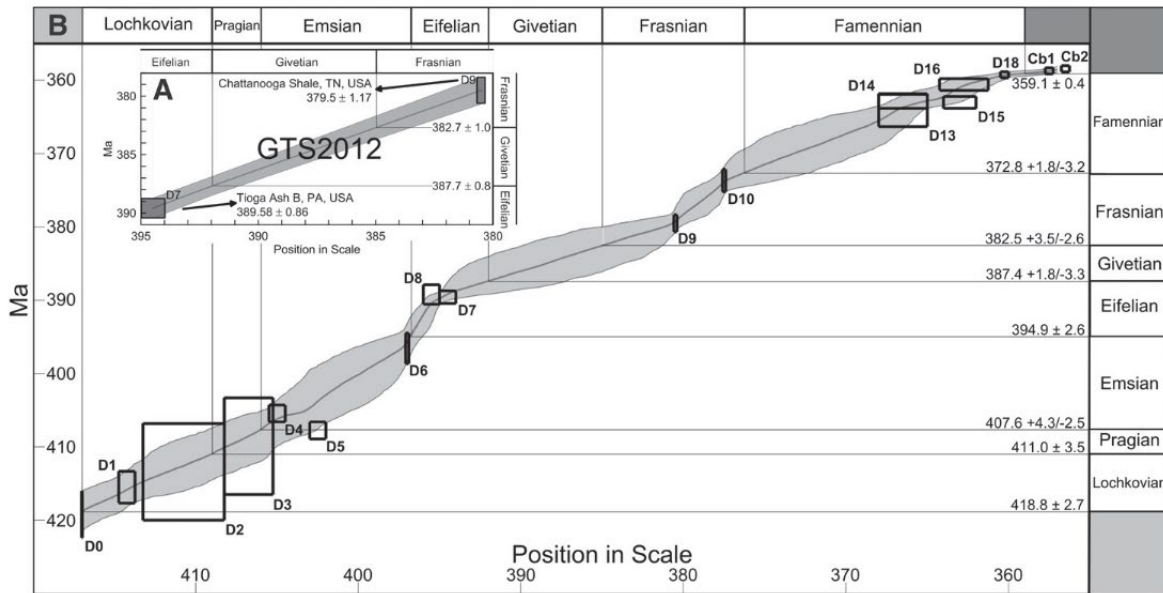
Limits of superposition



Time-averaging and stratigraphic disorder due to sediment mixing

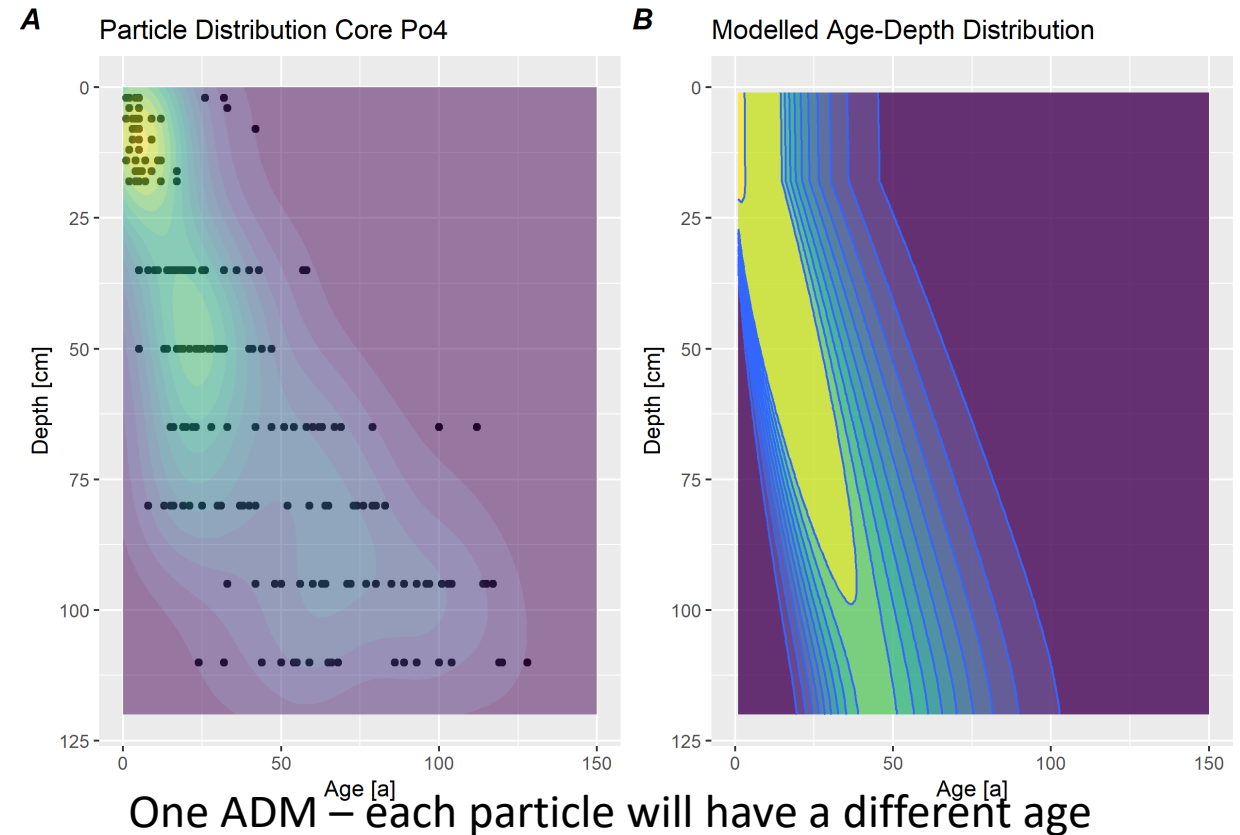
Dominant driver:
Physics/biology of particle deposition

Differences in uncertainty



Many possible ADMs – we're not sure which one it is

Infinite data & precision:
Will collapse to a line



One ADM – each particle will have a different age

Infinite data & precision:
Will converge to an age density

Levels of Age-depth models

Scale	Name	Units	E happened X years ago	E lasted X years	E1 is Z times longer than E2	E1 happened before E2	Strongest conclusion possible	Dominant source of information
Absolute	Absolute ADM	Ma	x	x	x	x	Absolute age	Geochronology
Interval	Anchored ADM	Myr		x	x	x	Absolute rates/fluxes	Sedimentary systems
Ratio	Floating/relative ADM	-			x	x	Relative rates/fluxes	Relative sed. rates
Ordinal	???	-				x	Ordering	Assumed superposition

Conclusion

- A wide range of questions can be answered without absolute ages
- This relies on understanding sedimentary systems

Discuss:

Which systems/timescales are you working on?

Which type of age-depth model is most relevant for you question?